Roadmap. To compute correlation functions order by order, $Z(J)$ of φ^3 theory is expanded into Feynman diagrams, each weighted by $1/S$. Then $Z = e^{iW}$: iW sums the connected diagrams, the vacuum bubbles are normalized away, and Y, fixed by $\langle 0|\varphi|0\rangle = 0$, cancels every tadpole. Used next: LecNote9 (§10), Feynman rules and $2 \to 2$; LecNote12–13 (§14, §16) fix A, B, Z_g. Supplements: SymmetryFactors, AsymptoticStates, AsymptoticSeries.

Sec. 9. Path Integral for Interacting theory

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial_\mu \varphi \partial^\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 + \frac{1}{6} Z_g g \varphi^3 + Y\varphi$$

$\boxed{m}$ is identified as particle mass

$\boxed{g}$ is coupling strength whose meaning will be more clearly defined later.

The Z's and Y are defined such that

$\boxed{\langle 0|\varphi(x)|0\rangle = 0}$ and $\boxed{\langle k|\varphi(x)|0\rangle = e^{-ipx}}$

$\langle 0|0\rangle = 1$ and $|k\rangle$ one-particle state with four momentum k^μ ($k^2 = -m^2$)

such that

$$\langle k'|k\rangle = (2\pi)^3 2k^0 \delta^3(\vec{k} - \vec{k}')$$

These four conditions $\square$ determines three Z's and Y.

This theory is called φ^3 theory.

(this theory is not bounded from below, but we just use it to demonstrate perturbation theory; ignoring the stumbling process)

(more discussion can be done after class)

Also comment on φ^4 theory.

We want to evaluate

$$Z(J) \equiv \langle 0|0\rangle_J = \int \mathcal{D}\varphi \, e^{i\int d^4x \,[\mathcal{L}_0 + \mathcal{L}_1 + J\varphi]}$$

Similar to previously learned in the Path Integral formalism for Quantum Mechanics, we have

$$Z(J) = e^{i\int d^4x \,\mathcal{L}_1\left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)} \int \mathcal{D}\varphi \, e^{i\int d^4x \,[\mathcal{L}_0 + J\varphi]}$$

$$\propto e^{i\int d^4x \,\mathcal{L}_1\left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)} Z_0(J) \quad \text{--- Eq(9.6)}$$

where $Z_0(J)$ is for the free theory

$$Z_0(J) = e^{\frac{i}{2}\int d^4x\, d^4x' \, J(x)\Delta(x-x')J(x')}$$

the 1-$i\epsilon$ treatment gives additional factors so that we need to define / require that $Z(0)=1$ ← normalization of probability

In the above equation, we implicitly assumed that

$$\mathcal{L}_0 = -\frac{1}{2}\partial^\mu\varphi\,\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2$$

the rest needs to be included in $\mathcal{L}_1$.

$$\mathcal{L}_1 = \frac{1}{6}Z_g g\varphi^3 + \cancel{\frac{1}{2}Z_\varphi \varphi^2}\cdot + \mathcal{L}_{CT}$$

$$\mathcal{L}_{CT} = -\frac{1}{2}(Z_\varphi - 1)\partial^\mu\varphi\,\partial_\mu\varphi - \frac{1}{2}(Z_m - 1)m^2\varphi^2 + \gamma\varphi.$$

↑ counter-term

One anticipate $g \to 0$, then $I_1 = 0$

so $Y \underset{g\to 0}{\to} 0$, $Z_i \underset{g\to 0}{\to} 1$

we shall see that $Y = O(g)$, $Z_i = 1 + O(g^2)$

To compute Eq (9.6), a lot of functional derivatives are needed $I_1\left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)$.

Let's focus on the "leading-order" result

$$Z_1(J) \propto e^{\left[\frac{i}{6} Z_g g \int d^4x \left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)^3\right]} Z_0(J).$$

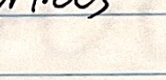
↑ leading order

↺ 1st order perturbation [$Z_1(0) = 1$ required].

Expanding the exponential functions

$$Z_1(J) \propto \sum_{V=0}^{\infty} \frac{1}{V!}\left[\frac{iZ_g g}{6}\int d^4x \left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)^3\right]^V$$

$$\times \sum_{P=0}^{\infty} \frac{1}{P!}\left[\frac{i}{2}\int d^4y\, d^4z\, J(y)\Delta(y-z)J(z)\right]^P.$$

V counts # of vertices 

P counts # of propagators

The number of external (surviving sources)

$E = 2P - 3V$

each ϕ provide 2 J's
each V get rid of 3 J's

for a fixed order in ϕ and V expansion.

These can be organized via the Feynman diagrams.

each line segment is $\frac{1}{i}\Delta(y-z)$

each filled circle is a source $i\int d^4x J(x)$ [1]

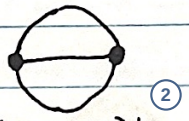
each vertex joining three lines is $i Z_g g \int d^4x$
for ϕ^3 theory

[1] In these drawings a source is a hatched oval (p. 5 on); the small filled dots are vertices.

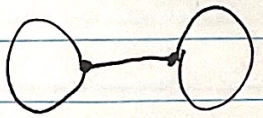
We can draw various diagrams.

[2] S counts the relabelings that map a term to itself. The $V!\,6^V P!\,2^P = 3456$ relabelings (vertices, legs, lines, line ends) match the prefactor, but $2! \times 3!$ of them (swap vertices, permute lines) fix each theta term. Of the $6! = 720$ ways the six $\delta/\delta J$ hit the six J's, 288 give the theta: $288/3456 = 1/12$.

e.g. $E=0$, $V=2$
$(P=3)$

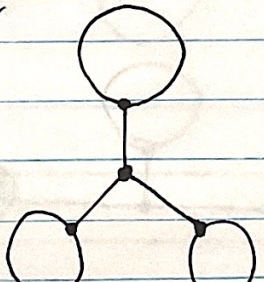


$S = 2\times 3!$ [2]



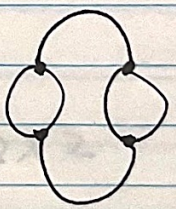
$S = 2\times 2\times 2$

$E=0$, $V=4$
$(P=6)$



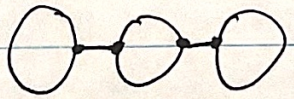
$S = 3!\times 2^3$

$S = 4!$

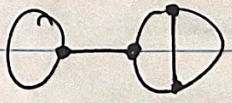


$S = 2^3\times 2$

But calculate it, we only have one possibility! How can I count it $2!\times 3!$ times?



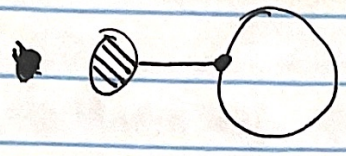
$S = 2^4$



$S = 2\times 2\times 2$

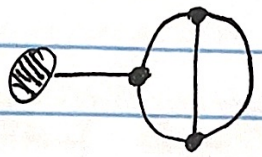
0 1 —

$E=1$ $V=1$
$(P=2)$

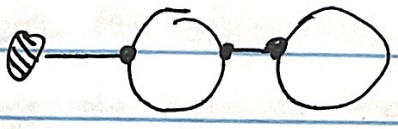


$S=2$

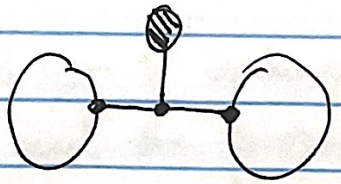
$E=1$ $V=3$
$(P=5)$



$S=2\times2$



$S=2\times2$

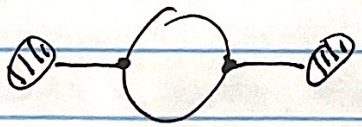


$S=2\times2^2$

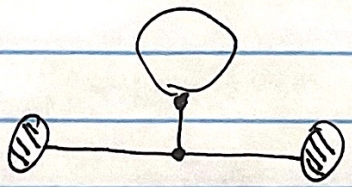
$E=2$, $V=0$
$(P=1)$

$S=2$

$E=2$, $V=2$
$(P=4)$



$S=2\times2$



$S=2\times2$

The essence is a diagrammatic counting of functional derivatives with the generating function Z_0.

we do symmetry factor counting for a reason that will be transparent later.

They represent Mathematical operations of functional derivatives and the Green's functions.

The three functional derivatives can be rearranged 3! without changing the diagrams.

We can rearrange the vertices themselves with V!

The two sources can be exchanged connecting the propagators by 2!

and the propagators themselves by P!

All together, they cancel explicitly the $\frac{1}{V!}\left(\frac{1}{6}\right)^V \cdot \frac{1}{P!}\left(\frac{1}{2}\right)^P$ factors in the dual Taylor expansion.

But this is an Overcounting, as different ways of moving vertices and propagators may yield a same diagram.

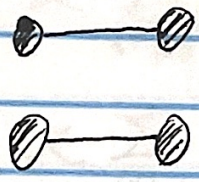
There are some symmetry properties of the diagrams

We can evaluate the overcounting by dividing the "symmetry" factors" S for each diagram. (Automorphism)

All the above connected diagrams are "connected"

There are others disconnected contributions to $Z(J)$ as well

e.g. for 4E



also contributes

Let C_I for a particular connected diagram

A general diagram can be expressed as

$$D = \frac{1}{S_D} \prod_I (C_I)^{n_I}$$

n_I counts # of C_I in D.

S_D is [additional] symmetry in D.

$S_D = \prod_I \frac{1}{n_I!}$ as exchanging identical C_Is do not generate new diagrams and each C_I is different.

1 Should read $S_D = \prod_I n_I!$ (e.g. two thetas: $S = 2! \cdot 12^2 = 288$), so that $D = \prod_I \frac{1}{n_I!}(C_I)^{n_I}$, as used on p. 8.

Now $Z_1[J]$ is given by summing all diagrams D, and each D is labeled by the integers η_I, (I is index of $\{\eta_I, C_I\}$)

$$Z_1[J] \propto \sum_{\{\eta_I\}} D$$

$$\propto \sum_{\{\eta_I\}} \prod_I \frac{1}{\eta_I!} (C_I)^{\eta_I}$$

$$\propto \prod_I \sum_{\eta_I=0}^{\infty} \frac{1}{\eta_I!} (C_I)^{\eta_I}$$

$$\propto \prod_I e^{C_I}$$

$$\propto \exp\left[\sum_I C_I\right]$$

A remarkable result!

$Z_1[J]$ is given by the **exponetial** of all **connected diagrams**

And to impose the normalization condition $Z_1[0] = 1$ we only need to require all the vacuum diagrams being zero. ($E=0$). Hence $Z_1[0] = \langle 0|0\rangle_{J=0} = 1 = \exp[C_0] = 0$.[1]

1 Should read $Z_1(0) = \mathcal{N} e^{C_0} = 1$, $C_0 \equiv \sum_{E=0} C_I$, $\mathcal{N}$ the constant in $\propto$. Vacuum diagrams are not zero: $\mathcal{N} = e^{-C_0}$ divides them out, hence $I \neq \{0\}$ on p. 9.

Now we have

$$Z_1[J] = \exp[iW_1[J]]$$

where

$$iW_1[J] = \sum_{I \neq \{0\}} C_I .$$ (1)

1 Each connected vacuum diagram carries a free $\int d^4x$, a factor VT: $C_0 = -i\,\Delta E_0\,T$, the vacuum-energy shift due to g (cf. LecNote3's E_0), a mere phase. It is J-independent, so it cancels in every correlator, $\langle 0|T\varphi(x_1)\cdots|0\rangle = Z(0)^{-1}\frac{\delta}{i\delta J(x_1)}\cdots Z\big|_{J=0}$.

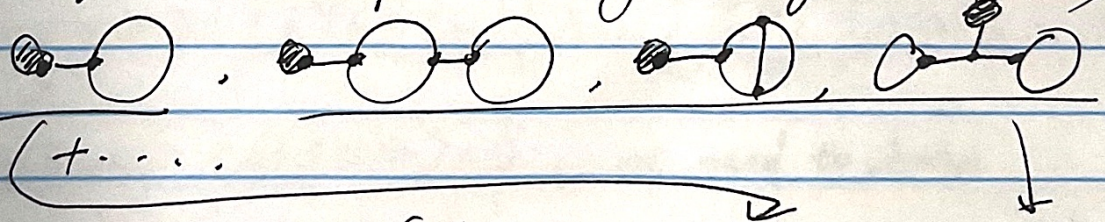
Recall that we wanted $Z[J]$ but we calculated $Z_1[J]$, which misses the counter terms that were introduced earlier for consistency.

Let's check the consequence.

$$\langle 0|\varphi(x)|0\rangle = \frac{1}{i}\frac{\delta}{\delta J(x)} Z_1[J]\Big|_{J=0}$$

$$= \frac{\delta}{\delta J(x)} W_1[J]\Big|_{J=0}$$

This is the sum of all diagrams with a single source (after removing remaining sources by $J=0$)



$$\langle 0|\varphi(x)|0\rangle = \frac{1}{2} ig \int d^4y \, \frac{1}{i}\Delta(x-y)\frac{1}{i}\Delta(y-y) + O(g^3).$$

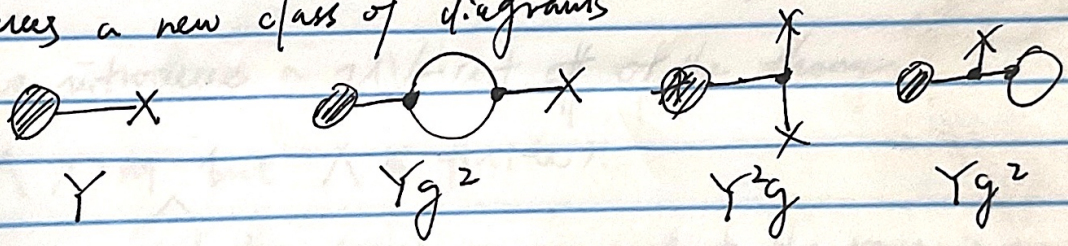
[1] LSZ needs $a_1^\dagger(\pm\infty)|0\rangle$ to be one particle, but for $v = \langle 0|\varphi|0\rangle \neq 0$, $\langle 0|a_1^\dagger(t)|0\rangle = (2\pi)^3 m\, v\, f_1(0)\, e^{-imt}$ never decays (AsymptoticStates). Y does LecNote4's shift $\varphi \to \varphi - v$ inside $\mathcal{L}$, so we expand about the true vacuum. Without it, tadpoles on lines also shift the mass: $m^2 \to m^2 - gv$.

Here we set $Z_g = 1$ for consistent counting of g. (perturbation implied) as $Z_g = 1 + O(g^2)$ ↑ to be proven later.

We see
$\langle 0|\varphi(x)|0\rangle \neq 0$ at order g.

we must introduce the counter term to enable the LSZ [1]

such a counter term $Y\varphi(x)$
introduces a new class of diagrams



assuming/recall Y is $O(g)$, we see the leading diagram is at order g.

so
$$\langle 0|\varphi(x)|0\rangle = \left(iY + \frac{1}{2}(ig)\frac{1}{i}\Delta(0)\right)\int d^4y \frac{1}{i}\Delta(x-y) + O(g^3)$$

so to have $\langle 0|\varphi(x)|0\rangle = 0$, we need to choose

$$Y = \frac{1}{2} ig\Delta(0) + O(g^3).$$

Y is supposed to be real, let's check

$\Delta(0)$ needs to be imaginary

$$\Delta(0)=\int\frac{d^4k}{(2\pi)^4}\frac{1}{k^2+m^2-i\epsilon}$$

this is also a divergent ~~di~~ integral

that there are $k^2=-m^2$ and infinite length of $|\vec{k}|$.

Again we encountered **Ultraviolate Divergence**

UV-divergence.

so we introduce a UV-cut off of the theory.

($\Lambda >> m$ but Λ finite).

and other scales we can conduct the experiment)

The UV-cut off of the propagator is introduced in the following subtitle way to respect Lorentz Invariance

$$\Delta(x-y)\to\int\frac{d^4k}{(2\pi)^4}\frac{e^{-ik(x-y)}}{k^2+m^2-i\epsilon}\left(\frac{\Lambda^2}{k^2+\Lambda^2-i\epsilon}\right)^2$$

for $\Lambda^2 >> m^2$ ~~no~~ for $k^2 << \Lambda^2$

it is dominated by the original pole but

for larger Λ k^2, it's Λ^2

now

$$\Delta(0) = \frac{i}{16\pi^2}\Lambda^2 \cdot$$ (1)

1 Why i: $\frac{1}{i}\Delta(0) = \langle 0|\varphi(x)^2|0\rangle = \int \frac{d^3k}{(2\pi)^3\, 2\omega} > 0$ (LecNote7), the vacuum fluctuation of the field at a point. Every mode adds $1/2\omega$, so it grows like Λ^2. So $Y = -\frac{g}{2}\langle\varphi^2\rangle \simeq -g\Lambda^2/32\pi^2$ is real: it cancels the linear piece of $\frac{g}{6}\varphi^3 = \frac{g}{6}{:}\varphi^3{:} + \frac{g}{2}\langle\varphi^2\rangle\varphi$.

we could formally take $\Lambda \to \infty$ with

$\langle 0|\varphi(x)|0\rangle = 0$.

A view point can be that

$Z_*(J)$ contains infinity part

but physical observables are the properly normalized probability $\left(\left|\frac{\langle f|i\rangle}{Z(J)}\right|^2_{J=0}\right)$, although (2)

both numerator and denominator contains infinity, but their ratio is finite.

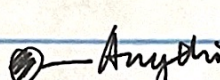
2 Should read $|\langle f|i\rangle|^2/|Z(0)|^2$. This ratio removes only the vacuum bubbles (p. 8); UV divergences of connected diagrams, like the Λ^2 above, remain until counterterms (Y here) cancel them.

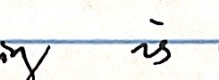
This is a remarkable simplification, meaning the sum of all connected diagrams with a single source is zero. (No matter which order)

We only need to calculate diagrams with more sources.

We will discuss the renormalizing counter terms later.

Note that



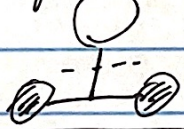
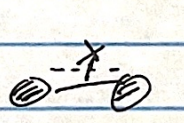
is canceled by


Then Any diagram that contains

Any diagram contains : $\frac{1}{i}$ Anything (no source) would be cancelled.

~~then~~ this Anything no source with one external line is called Tadpole terms.

e.g. for two external sources., ~~there is~~ there is no contribution from

because


cancels it

Tadpole terms do not contribute

Then the remaining $\mathcal{L}_{CT}$ are Z_φ and Z_m terms

$$Z(J) = e^{-\frac{i}{2}\int d^4x \left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)\left(-A\partial_x^2 + Bm^2\right)\left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)} Z_\varphi(J)$$

$$Z_\varphi \partial_\mu \varphi \partial^\mu \varphi \xrightarrow{\text{i.b.p.}} - Z_\varphi \varphi \partial^2 \varphi.$$ [1]

We can proceed and calculate to get the full $Z(J)$ [2]

1 Should read *by parts*, not E.O.M. Here $A \equiv Z_\varphi - 1$ and $B \equiv Z_m - 1$ ($\mathcal{L}_{\rm CT}$, p. 2).

2 The trailing factor is $Z_1(J)$ of p. 3 with all tadpoles dropped (they cancel against Y). So again $Z(J) = e^{iW(J)}$: iW = connected diagrams with $E \geq 2$ and no tadpoles, plus a two-line vertex $-i(Ak^2 + Bm^2)$ (LecNote9); $A, B = O(g^2)$ are fixed in LecNote12.